Confusion Matrix

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

TEACHER GUIDE · 1 OF 9

Confusion Matrix

> Prepare Confusion Matrix: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Build a labelled **confusion matrix** from matched reference and **prediction** pairs

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/13, ai-4/15. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Scores and Calibration Intuition

Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

Look for: Interpret calibration using groups of forecasts

Reference example, not the only acceptable wording: Teacher explanation: 2/10=.2 and 6/10=.6.

Nearest Neighbors

Without opening the old lesson, explain one key idea from Nearest Neighbors. Give an example and a mistake that the idea helps you avoid.

Look for: Compute absolute distances and rank **training** neighbours

Reference example, not the only acceptable wording: Distances to A,B,C,D,E are 1,1,5,7,9 cm. A and B are equally far from 3 on the numeric feature; their labels do not enter the distance calculation.

Dataset Cards

Without opening the old lesson, explain one key idea from Dataset Cards. Give an example and a mistake that the idea helps you avoid.

Look for: Inspect row meaning missing values and repeated records

Reference example, not the only acceptable wording: One row is intended to represent one indoor practice trip.

The source is a teacher-authored fictional export for this lesson, duration uses minutes, and None means unknown rather than zero.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Positive=spotted. TP=6,FN=2,FP=3,TN=9. Draw a matrix with actual labels in rows and **prediction**s in columns. Compute total and accuracy.

Concept to explain

A **confusion matrix** separates **true positive**s, **false positive**s, **false negative**s, and **true negative**s so error types remain visible.

Term	Meaning
true positive	positive case predicted positive
false positive	negative case predicted positive
false negative	positive case predicted negative
true negative	negative case predicted negative

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow

> extra sessions for the investigation, independent work and project.
>
> Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. Start with ID cards and a physical two-by-two table so students count matched pairs before learning abbreviations. Use blocked as positive consistently.

Keep raw counts visible and postpone precision and recall formulas to later weeks.

Readiness answer

IDs preserve which reference belongs to each **prediction** and which examples were wrong. The same accuracy can come from different **prediction**s and error patterns.

Printed lesson and workbook

Eight ID cards carrying the worked reference and **prediction** pairs

Two-by-two sorting mat labelled reference rows and **prediction** columns

Optional Python 3 editor with no packages required

End goal for the pictorial case

By the end, I can explain **confusion matrix** using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Confusion Matrix using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

Question to explain

QUESTION Construct all four cells from twenty **prediction**s and verify the total.

MODEL IDEA A **confusion matrix** separates **true positive**s, **false positive**s, **false negative**s, and **true negative**s so error types remain visible.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD For spotted=positive: TP=6, FP=3, FN=2, TN=9. Check total: 6 + 3 + 2 + 9 = 20 cases.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Step 1

STEP 1 For spotted=positive: TP=6, FP=3, FN=2, TN=9.

Step 2

STEP 2 Check total: 6 + 3 + 2 + 9 = 20 cases.

Step 3

STEP 3 Correct **prediction**s = TP + TN = 15; accuracy = 15 ÷ 20 = 75%.

Step 4

STEP 4 Inspect false alarms and misses separately before choosing an improvement.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

The teaching case

Positive is spotted. TP=8, FP=2, FN=4, TN=6.

The teaching case

Build actual-row/predicted-column matrix and total.

Exact explanation to model

Why the result follows

Actual spotted row is 8 spotted,4 healthy; actual healthy row is 2 spotted,6 healthy. Total=20; correct=14. State axes and **positive class**.

Explain the added worked example

Explain the added worked example · 1

A **confusion matrix** shows which labels a classifier mixes up. It counts examples after **prediction**s are made; it does not fit or repair the classifier.

Week 12 had reference/**prediction** pairs clear/clear, blocked/blocked, clear/blocked. A labelled table can preserve the type of that error as well as the total score.

Explain the added worked example · 2

This week uses eight new fictional evaluation records from a frozen classifier. They are not extra **prediction**s from the Week 12 code. Each ID appears once with a reviewed reference and one **prediction**.

The grouped table below lists every ID. Keep reference and **prediction** paired by ID before counting.

Explain the added worked example · 3

Use reference labels for rows and predicted labels for columns. On both axes the order is clear, then blocked. Declare blocked as the **positive class** and clear as negative.

Positive means the condition of interest, not good or safe. A missed blocked case and a false alarm are different errors.

Explain the added worked example · 4

index maps clear to 0 and blocked to 1. Each paired record adds one to its reference row and **prediction** column. The script prints [[2, 1], [2, 3]] then 5 8. It assumes unique IDs and only these two labels.

Check IDs and missing or unresolved references before counting; the code does not validate all those conditions.

The diagonal cells count correct **prediction**s: 2+3=5. Off-diagonal cells count mistakes: 1+2=3. Every record must contribute exactly once. This is a count matrix, not percentages or probabilities.

Keep the axis labels even when copying the numbers.

Key terms

Term	Meaning
Confusion matrix	A table counting reference and prediction combinations
Positive class	The class chosen as the condition of interest
False positive	A positive prediction when the reference is negative
False negative	A negative prediction when the reference is positive

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Blocked is the positive class

REFERENCE AND PREDICTION	CELL NAME	IDS AND COUNT
clear then clear	True negative TN	V1 V2 count 2
clear then blocked	False positive FP	V3 count 1
blocked then clear	False negative FN	V4 V5 count 2
blocked then blocked	True positive TP	V6 V7 V8 count 3

Read reference first and prediction second. Here a false positive is a false blocked alarm; a false negative is a missed blocked case.

Place V3 and V4 in the matrix. Explain which is a false positive and which is a false negative when blocked is positive.

True means **prediction** agrees with the reference. It does not prove that the reference annotation is infallible. Resolve disputed labels with the documented review policy rather than silently changing them to match **prediction**s.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?

Thinking move	Teacher prompt
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Confusion Matrix

Read the **confusion matrix** as four distinct stories about a plant decision.

The facts we will use

> **Supplied facts**
>
> Positive means spotted. Toy results: TP=4, FN=1, FP=3, TN=12. Rows are actual spotted/healthy; columns predicted spotted/healthy.

> **Supplied facts**
>
> All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/15

Positive means spotted

	predicted spotted	predicted healthy
actual spotted	4	1
actual healthy	3	12

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/15

Confusion Matrix

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Fill actual spotted row

Actual spotted: [4, 1]. Four caught; one missed.

2. Fill actual healthy row

Actual healthy: [3, 12]. Three false alarms; twelve correctly clear.

3. Compute accuracy

Correct=4+12=16 of total 20, giving80%.

CHANGE ONE THING

A review corrects one missed spotted leaf, moving it from FN to TP.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Fill actual spotted row	Actual spotted: [4, 1]. Four caught; one missed.
2. Fill actual healthy row	Actual healthy: [3, 12]. Three false alarms; twelve correctly clear.
3. Compute accuracy	Correct=4+12=16 of total 20, giving80%.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Correct=4+12=16 of total 20, giving**80%**. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

The matrix depends on positive-class definition and reference labels. Accuracy alone hides the difference between misses and false alarms.

Change one thing in the pictured case

Change one thing

Change: A review corrects one missed spotted leaf, moving it from FN to TP.

Predicted result

Prediction: Counts become TP 5, FN 0, FP 3, TN 12; accuracy is 17/20=85%.

Why the result follows

Reason: One decision becomes correct while the total number of evaluated leaves remains20.

Ask while showing the picture

Ask while showing the picture · 5

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Confusion Matrix, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Positive is spotted. TP=8, FP=2, FN=4, TN=6.

2. State the decision to explain

Write the question to explain: Build actual-row/predicted-column matrix and total.

3. Reveal the worked reasoning

Reveal after **prediction**s: Actual spotted row is 8 spotted,4 healthy; actual healthy row is 2 spotted,6 healthy. Total=20; correct=14. State axes and **positive class**.

Hold back the independent task

Keep for an independent attempt: Treat fern as positive. Counts are TP=6, FP=2, FN=3 and TN=9. How many examples and how many correct **prediction**s are there?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the

> result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: V3 belongs at reference clear/predicted blocked, FP. V4 belongs at reference blocked/predicted clear, FN. Exit: both have five diagonal matches, but distribute three mistakes differently: FP=1,FN=2 versus FP=0,FN=3.

Report identical evaluation records, axes, label order, **positive class**, totals and task consequences before comparing.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the three Week 12 reference/prediction pairs and explain why accuracy loses error detail.
7 to 18 minutes	Sort all eight new ID cards into labelled cells. Complete W1 before showing code output.
18 to 32 minutes	Trace the counting code. Connect each increment to a cell and check the diagonal.
32 to 45 minutes	Declare blocked positive, read the visual and complete W2 and the guided question.
Session 2 start to 8 minutes	Retrieve row versus column meaning and the positive class without relying on cell position alone.
8 to 20 minutes	Check totals and compare equal-accuracy matrices in W3. Explain the different error tradeoffs.
20 to 35 minutes	Collect W4 independently before support. Require labelled axes, totals and ID-level error explanations.
35 to 45 minutes	Use W5 and the exit question to limit claims. Offer the fresh retry after feedback and record support.

Reduce help deliberately

Completion practice

Completion task: Declare blocked positive. Give TP,TN,FP,FN counts and explain V3 and V4 in plain language. Does positive mean a desirable outcome?

Reasoning to listen for

Teacher check: TP=3, TN=2, FP=1, FN=2. V3 predicts blocked for a reference-clear case, a false alarm or FP. V4 predicts clear for a reference-blocked case, a missed blocked case or FN.

Positive names the chosen class; it does not mean desirable.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Confusion matrices. Proposed explanation: “Positive means good”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Positive is the chosen condition of interest; blocked is positive in this exercise.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why is adding TP alone without reducing FN an incorrect revision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

The same leaf changes category; it does not become a new evaluated case. Move one count from FN to TP to preserve the total 20. Changed-case check: Counts become TP 5, FN 0, FP 3, TN 12; accuracy is 17/20=85%.

One decision becomes correct while the total number of evaluated leaves remains20.

Changed-case reasoning: Counts become TP 5, FN 0, FP 3, TN 12; accuracy is 17/20=85%. One decision becomes correct while the total number of evaluated leaves remains20.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Confusion Matrix investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. Matrix Build: convert twenty actual/predicted pairs into four cells, recompute totals, and explain each error in plain language.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR The diagonal alone tells everything important about a classifier.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Positive means good: Positive is the chosen condition of interest; blocked is positive in this exercise.

A matrix has one universal visual orientation: Check whether rows are references or **prediction**s and inspect class order. Our convention uses reference rows and **prediction** columns.

Zero errors in a cell means no future risk: It is an observed count on one evaluation set. Coverage, sample size and future conditions still matter.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Confusion Matrix questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Actual spotted row is 8 spotted,4 healthy; actual healthy row is 2 spotted,6 healthy. Total=20; correct=14. State axes and **positive class**.

Independent question

Independent question · withhold the key

Treat fern as positive. Counts are TP=6, FP=2, FN=3 and TN=9. How many examples and how many correct **prediction**s are there?

Expected independent answer

Teacher key · Expected independent answer

Total is 6+2+3+9 = 20. Correct **prediction**s are TP+TN = 15; accuracy is 15/20 = 75%.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions

Criterion	0 points	1 point	2 points
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: "**False positive**s and **false negative**s belong in the correct-**prediction** total." Keep the first attempt and an uncoached retry separate.

Original transfer answer

> **Teacher key · Original transfer answer**
>
> Actual-positive row is 6 predicted-positive and 2 predicted-negative; actual-negative row is 3 and 9. Total=20, correct=15, accuracy=75%. Axes and class labels are essential; swapping FP and FN changes later metrics.

Explanations for the additional workbook examples

W1 Build the count matrix

> **Workbook question · W1**
>
> Use all eight worked IDs. Draw a matrix with reference rows and **prediction** columns, both ordered clear then blocked. Place the IDs in cells before writing the counts.

Teacher reasoning · W1

Reference clear/predicted clear contains V1,V2: 2. Clear/blocked contains V3: 1. Blocked/clear contains V4,V5: 2. Blocked/blocked contains V6,V7,V8: 3. Matrix [[2,1],[2,3]].

W2 Explain the two error types

Workbook question · W2

Declare blocked positive. Give TP,TN,FP,FN counts and explain V3 and V4 in plain language. Does positive mean a desirable outcome?

Teacher reasoning · W2

TP=3, TN=2, FP=1, FN=2. V3 predicts blocked for a reference-clear case, a false alarm or FP. V4 predicts clear for a reference-blocked case, a missed blocked case or FN.

Positive names the chosen class; it does not mean desirable.

W3 Audit totals and equal accuracy

Workbook question · W3

Give the worked row totals, column totals, total count and accuracy. Compare its errors with [[3,0],[3,2]] using the same axes and eight reference records.

Teacher reasoning · W3

Rows total 3,5; columns total 4,4; total 8; accuracy (2+3)/8=5/8=62.5 percent. The comparison matrix also scores 5/8, but has FP=0 and FN=3 rather than FP=1 and FN=2. Same accuracy can hide different error types.

W4 Count a fresh evaluation independently

Workbook question · W4

Each pair below is reference then **prediction**: R1 clear/clear, R2 clear/blocked, R3 clear/blocked, R4 blocked/blocked, R5 blocked/clear, R6 blocked/blocked. Use clear then blocked on both axes and reference rows.

Build the matrix, give row and column totals, name TP,TN,FP,FN with blocked positive, and calculate accuracy.

Teacher reasoning · W4

Matrix [[1,2],[1,2]]. Row totals 3,3; column totals 2,4; total 6. TP=2,TN=1,FP=2,FN=1. Accuracy (1+2)/6=3/6=50 percent. False alarms are R2,R3; the missed blocked case is R5.

W5 Interpret a zero count carefully

Workbook question · W5

A report gives zero **false positive**s but omits the evaluated sample size, axes and class definitions. What information is missing, and why does zero not prove that future false alarms are impossible?

Teacher reasoning · W5

Request split and sample size, labelled reference/**prediction** axes, class order and positive-class definition, plus the remaining counts.

Zero describes only the observed evaluation cases; limited coverage or few negative cases may hide future errors. It is not a guarantee.

Assess independent transfer

Independent question · withhold the key

Selected case: On20 cases: **true positive**s5, **false negative**s4, **false positive**s2, **true negative**s9. Build the **confusion matrix** with actual class as rows and predicted class as columns. Compute accuracy.

Selected case answer

Teacher key · Selected case answer

Actual positive row: predicted positive5, predicted negative4. Actual negative row: predicted positive2, predicted negative9. Total20; correct14; accuracy**70%**. Actual positives total9.

State the positive-class definition before interpreting the errors.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Confusion Matrix to Responsible Plant Classifier, adapt support and plan a later changed-case check.

Responsible Plant Classifier

Publish the Plant Classifier **confusion matrix** with positive-class definition, counts, denominator, and test-set version.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named **positive class**. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 16: Accuracy and Imbalance. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Give a blank labelled matrix and move one ID card at a time. Ask for reference first, **prediction** second before naming a cell. Extension: declare clear positive without changing the worked records.

Then TP=2,TN=3,FP=2,FN=1; explain why the original matrix counts and accuracy remain unchanged if axis order stays clear then blocked.

Require W4 [[1,2],[1,2]], row totals 3,3, column totals 2,4, TP=2,TN=1,FP=2,FN=1 and accuracy 3/6. Fresh retry: S1 blocked/blocked, S2 blocked/clear, S3 clear/clear, S4 clear/clear, S5 clear/blocked, always reference first.

Matrix [[2,1],[1,1]], rows 3,2, columns 3,2, TP=1,TN=2,FP=1,FN=1, accuracy 3/5. Require labelled axes and blocked positive before accepting abbreviations.

Week 14 covers accuracy and class imbalance. Carry forward row totals, class definitions and the full **confusion matrix**.

Teacher reference: https://scikit-learn.org/stable/modules/generated/sklearn.metrics.confusion_matrix.html (reference rows, **prediction** columns and label order; reviewed 2026-09-28).

Plan the delayed return

Suggested return points in this level: ai-4/16, ai-4/18, ai-4/21. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	

Record	Teacher evidence
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Confusion Matrix lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Build a labelled **confusion matrix** from matched reference and **prediction** pairs

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Declare blocked positive. Give TP,TN,FP,FN counts and explain V3 and V4 in plain language. Does positive mean a desirable outcome?

Supported practice answer

Reasoned practice answer: TP=3, TN=2, FP=1, FN=2. V3 predicts blocked for a reference-clear case, a false alarm or FP. V4 predicts clear for a reference-blocked case, a missed blocked case or FN.

Positive names the chosen class; it does not mean desirable.

Check every learner before discussion: Use the worked case for Confusion matrices. Proposed explanation: “Positive means good”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Positive is the chosen condition of interest; blocked is positive in this exercise.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: A review corrects one missed spotted leaf, moving it from FN to TP. Require a **prediction** and a reason before comparing. Expected result: Counts become TP 5, FN 0, FP 3, TN 12; accuracy is 17/20=85%.

Reason: One decision becomes correct while the total number of evaluated leaves remains20.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words

Criterion	Not yet	With support	Independent
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

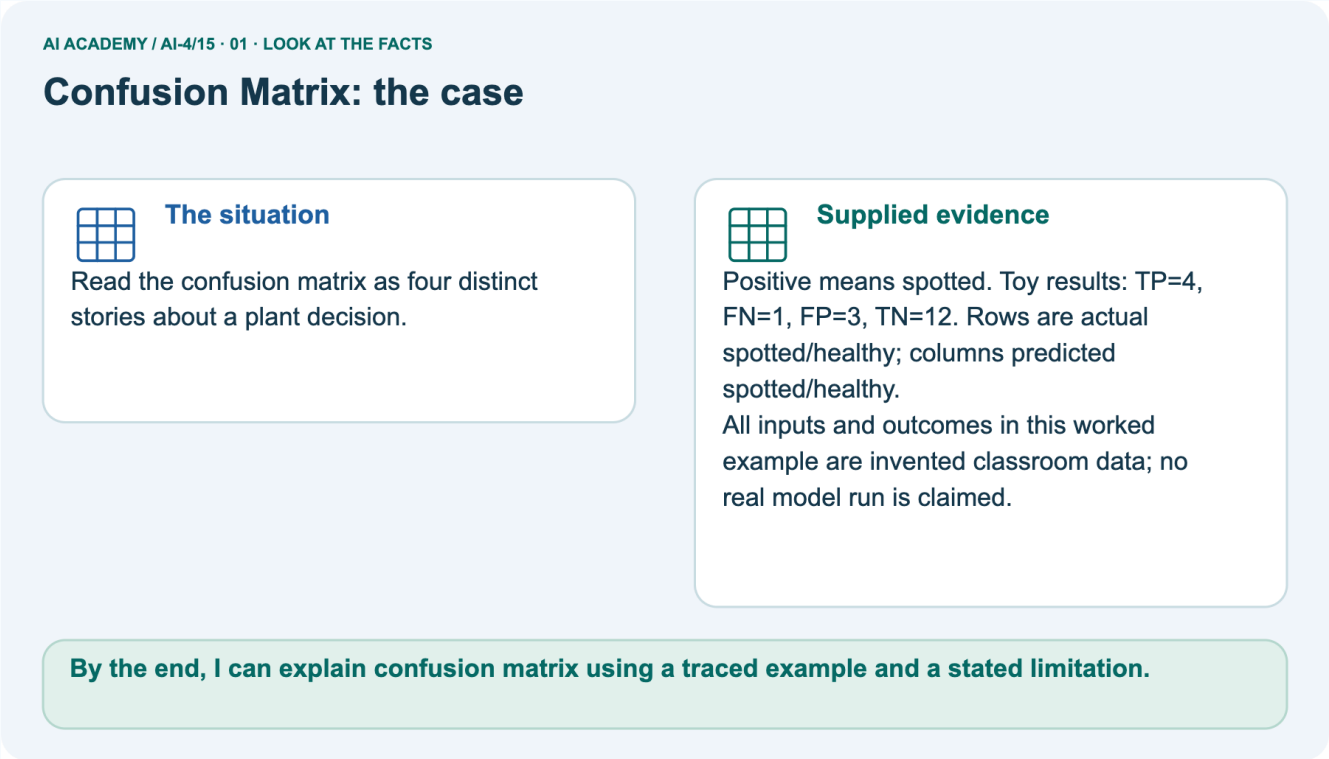
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Confusion Matrix graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Confusion Matrix: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the **confusion matrix** as four distinct stories about a plant decision. Supplied facts: Positive means spotted. Toy results: TP=4, FN=1, FP=3, TN=12. Rows are actual spotted/healthy; columns predicted spotted/healthy.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

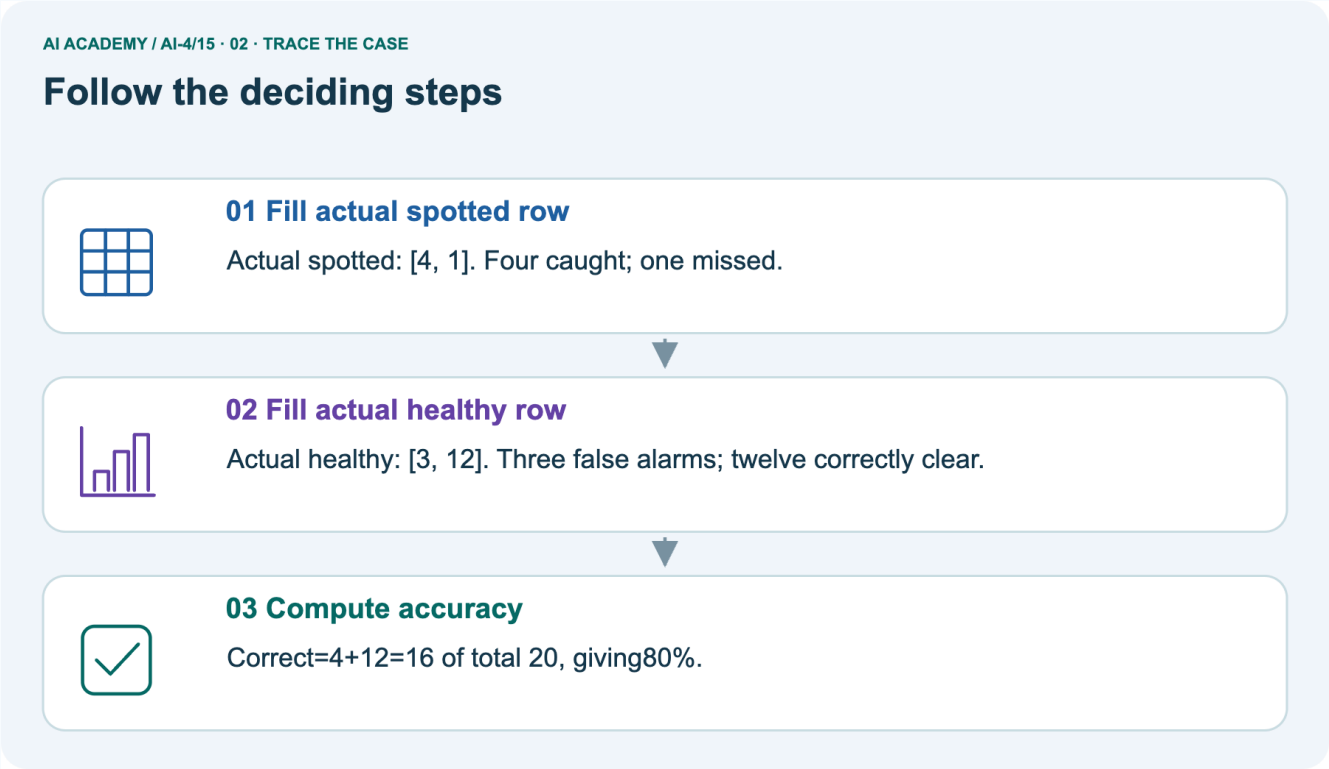
> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the

image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Fill actual spotted row: Actual spotted: [4, 1]. Four caught; one missed. → Fill actual healthy row: Actual healthy: [3, 12].

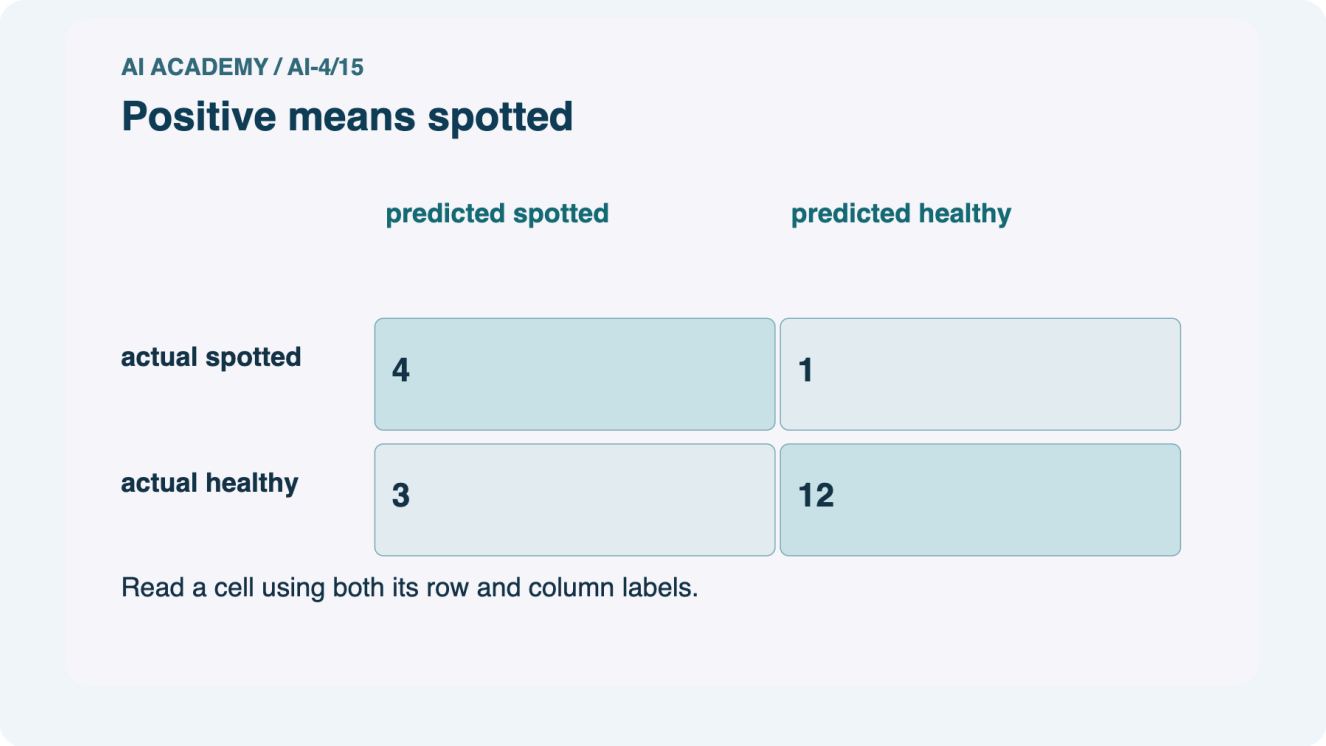
Three false alarms; twelve correctly clear. → Compute accuracy: Correct=4+12=16 of total 20, giving**80%**.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Positive means spotted



Follow the labelled picture with the matching explanation. Open the vector diagram

Positive means spotted. actual spotted: predicted spotted = 4, predicted healthy = 1. actual healthy: predicted spotted = 3, predicted healthy = 12.

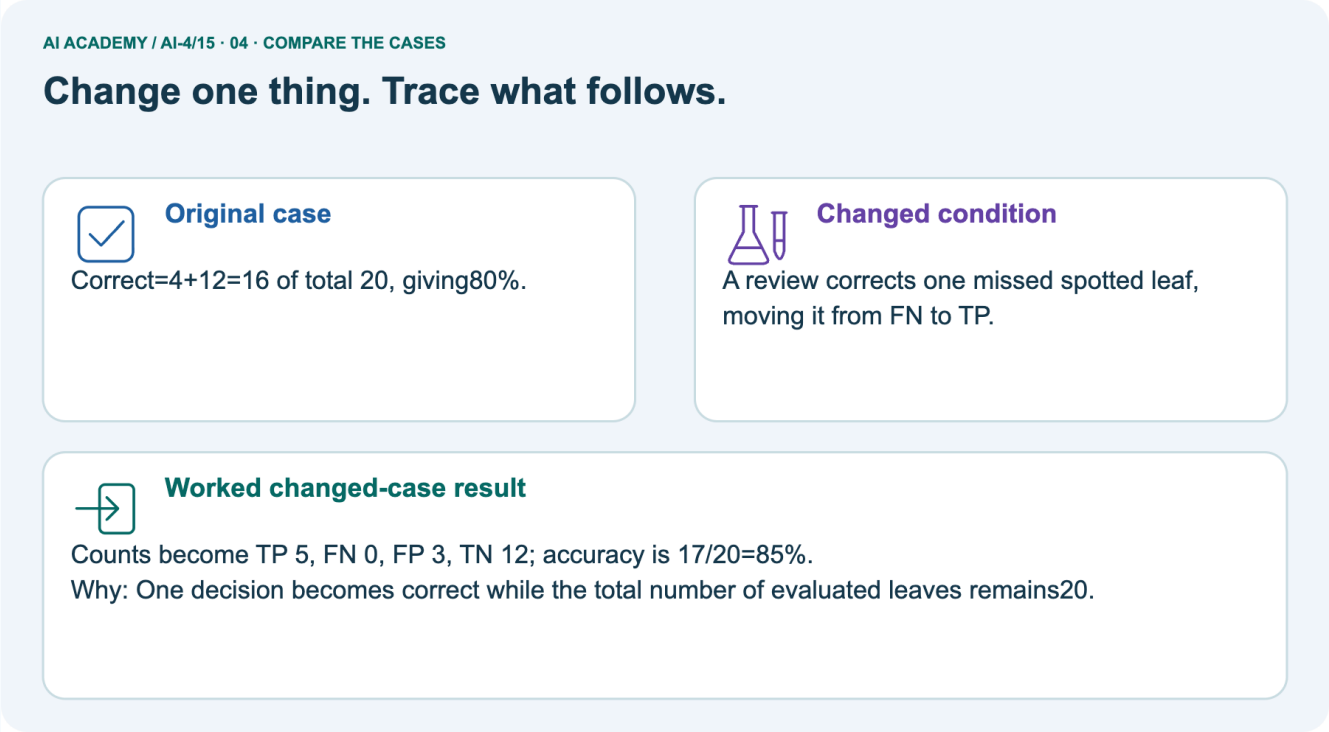
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Correct=4+12=16 of total 20, giving**80%**. Changed condition: A review corrects one missed spotted leaf, moving it from FN to TP. Expected result: Counts become TP 5, FN 0, FP 3, TN 12; accuracy is 17/20=85%.

Reason: One decision becomes correct while the total number of evaluated leaves remains20.

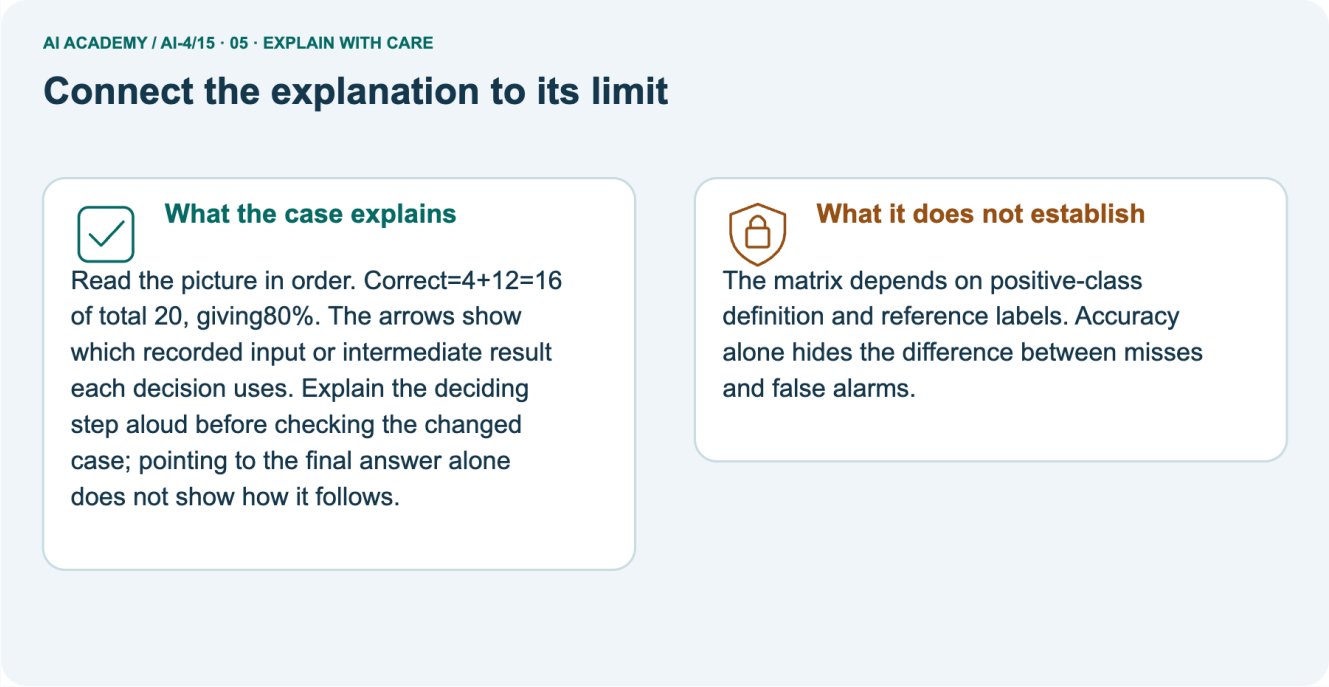
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: One decision becomes correct while the total number of evaluated leaves remains20.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Correct=4+12=16 of total 20, giving**80%**. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: The matrix depends on positive-class definition and reference labels.

Accuracy alone hides the difference between misses and false alarms.

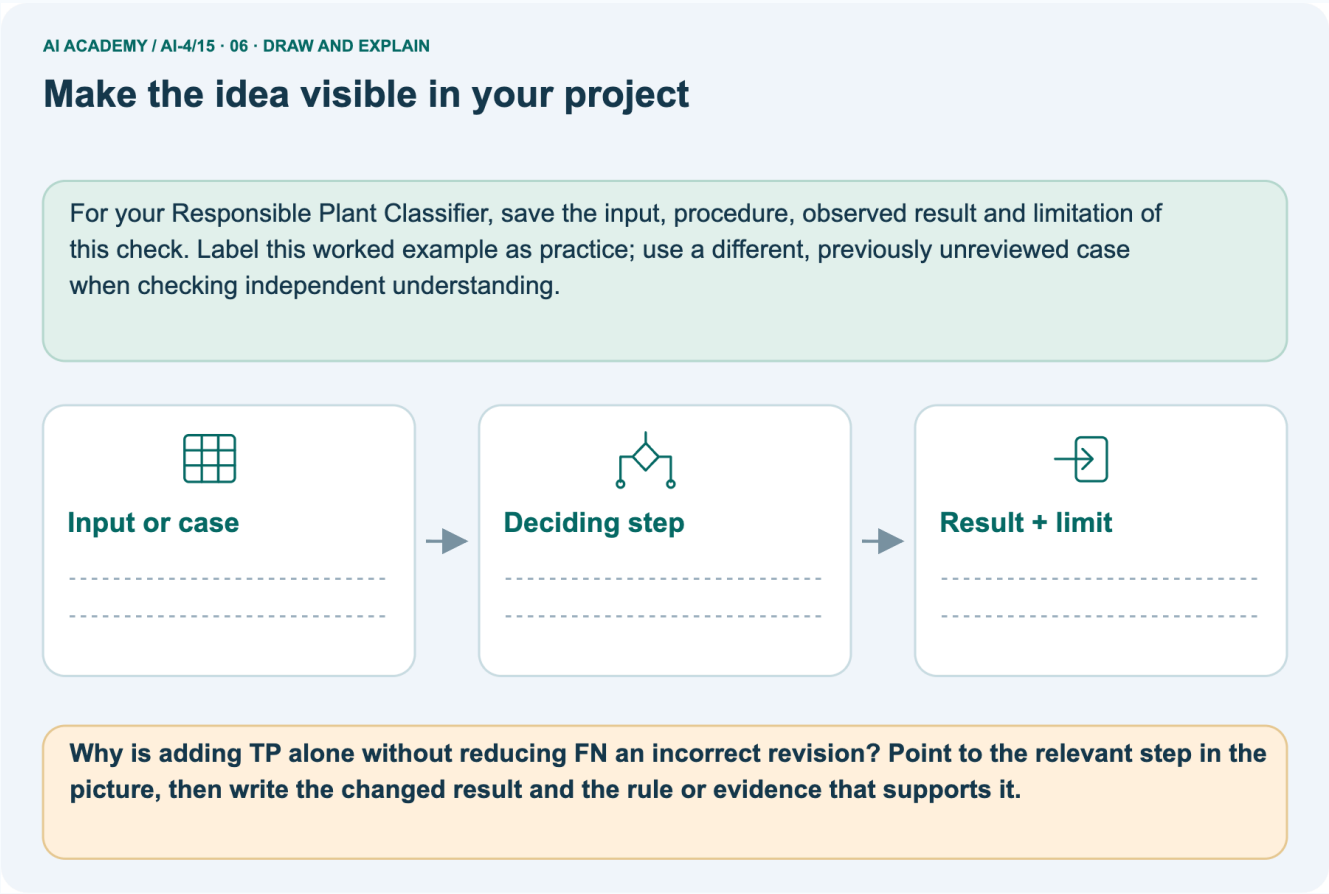
> Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: The matrix depends on positive-class definition and reference labels. Accuracy alone hides the difference between misses and false alarms.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is adding TP alone without reducing FN an incorrect revision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The same leaf changes category; it does not become a new evaluated case. Move one count from FN to TP to preserve the total 20.

Changed-case check: Counts become TP 5, FN 0, FP 3, TN 12; accuracy is 17/20=85%. One decision becomes correct while the total number of evaluated leaves remains20.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.